Student can explain how to implement resource requests and limits for pods and describe what happens if a pod exceeds its memory limit

Declined by Andrei Injagin

Accept Decline

Describe the issue

actual manifests for backend and frontend lack these definitions. Without them, the cluster cannot manage resources effectively

111 / 2000

Application manifests are successfully applied and deployed

Declined by Andrei Injagin

Accept Decline

Describe the issue

The CI/CD pipeline in .gitea/workflows/ci-cd.yml is configured to look for manifests in the k8s/ folder, but the files are actually in manifests/kubernetes/. This path mismatch prevents successful

190 / 2000

The CI/CD pipeline in .gitea/workflows/ci-cd.yml is configured to look for manifests in the k8s/ folder, but the files are actually in manifests/kubernetes/. This path mismatch prevents successful automated deployment.

Persistent Volumes (PVs) and Persistent Volume Claims (PVCs) are defined

Declined by Andrei Injagin

Accept Decline

Describe the issue

No PV or PVC manifests were found in the project. This is a mandatory requirement to ensure data (like logs or DB state) survives pod restarts.

118 / 2000

EFK stack is deployed and Fluentd/Fluent Bit is configured to collect logs

Declined by Andrei Injagin

Accept Decline

Describe the issue

Elasticsearch is deployed as a regular Deployment instead of a StatefulSet, and it lacks persistent storage. This setup is not durable and will lose all logs if the pod restarts.

149 / 2000

Alert is created for container memory usage exceeding 80% of its limit

`docker run -m 512m --name memory_test ubuntu /bin/bash -c "stress-ng --vm 1 --vm-bytes 450M --timeout 360s"` or similar to simulate high container memory usage. Adjust time as necessary.

Declined by Andrei Injagin

Accept Decline

Describe the issue

The alert rule exists, but since no memory limits are defined in the deployment manifests (see Test #10), this alert will never be triggered.

118 / 2000

Alert is created for Elasticsearch cluster status changing to yellow or red

Stop one of the Elasticsearch nodes in a multi-node cluster to simulate a yellow state

Declined by Andrei Injagin

Accept Decline

Describe the issue

This specific alert rule is missing from the prometheus-rules.yaml file, leaving the logging system unmonitored.

98 / 2000

Alert is created for Fluentd log collection errors

Misconfigure Fluentd to simulate log collection errors, e.g., by providing incorrect log source paths

Declined by Andrei Injagin

Accept Decline

Describe the issue

Similar to Test #44, this required alert was not implemented in the Prometheus configuration.

80 / 2000